ECE2002
DIGITAL LOGIC DESIGN

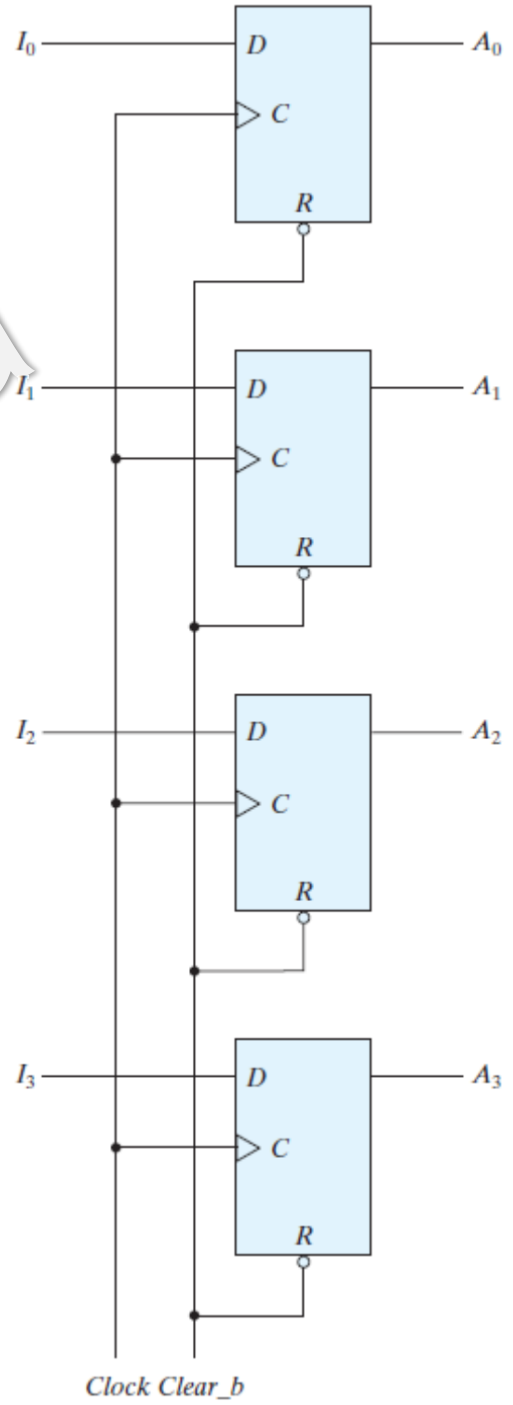
Module 4 – Lecture 1

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Registers

- Group of flip-flops sharing a common clock and is capable of storing one bit of information each.
- An *n*-bit register consists of a group of *n* flip-flops capable of storing *n* bits of binary information.
- In addition to the flip-flops, a register may have combinational gates that perform certain data-processing tasks.
- The figure to the right is a register constructed with four *D*-type flip-flops to form a four-bit data storage register.
- The common clock input triggers all flip-flops on the positive edge of each pulse, and the binary data available at the four inputs are transferred into the register.
- The value of (I_3, I_2, I_1, I_0) immediately before the clock edge determines the value of (A_3, A_2, A_1, A_0) after the clock edge.
- The input *Clear_b* goes to the active-low *R* (reset) input of all four flip-flops. When this input goes to 0, all flip-flops are reset asynchronously. The *R* inputs must be maintained at logic 1 (i.e., de-asserted) during normal clocked operation.

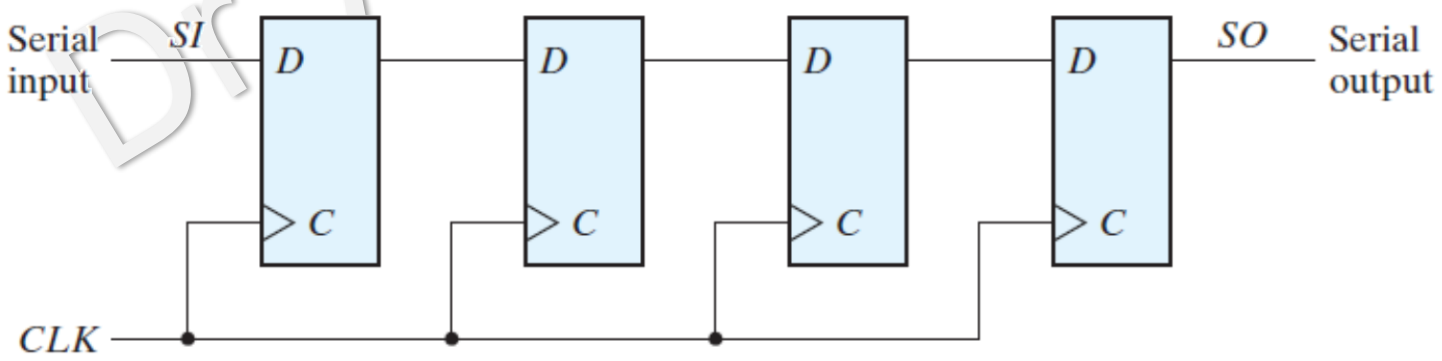


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Registers

Shift Register

- A register capable of shifting the binary information held in each cell to its neighboring cell in a selected direction.
- It consists of a chain of flip-flops in cascade, with the output of one flip-flop connected to the input of the next flip-flop.
- All flip-flops receive common clock pulses, which activate the shift of data from one stage to the next.
- Can be unidirectional (left shift or right shift) or bidirectional (both left and right shifts)
- This 4-bit shift register below is unidirectional (right shift). Each clock pulse shifts the contents of the register one bit position to the right.

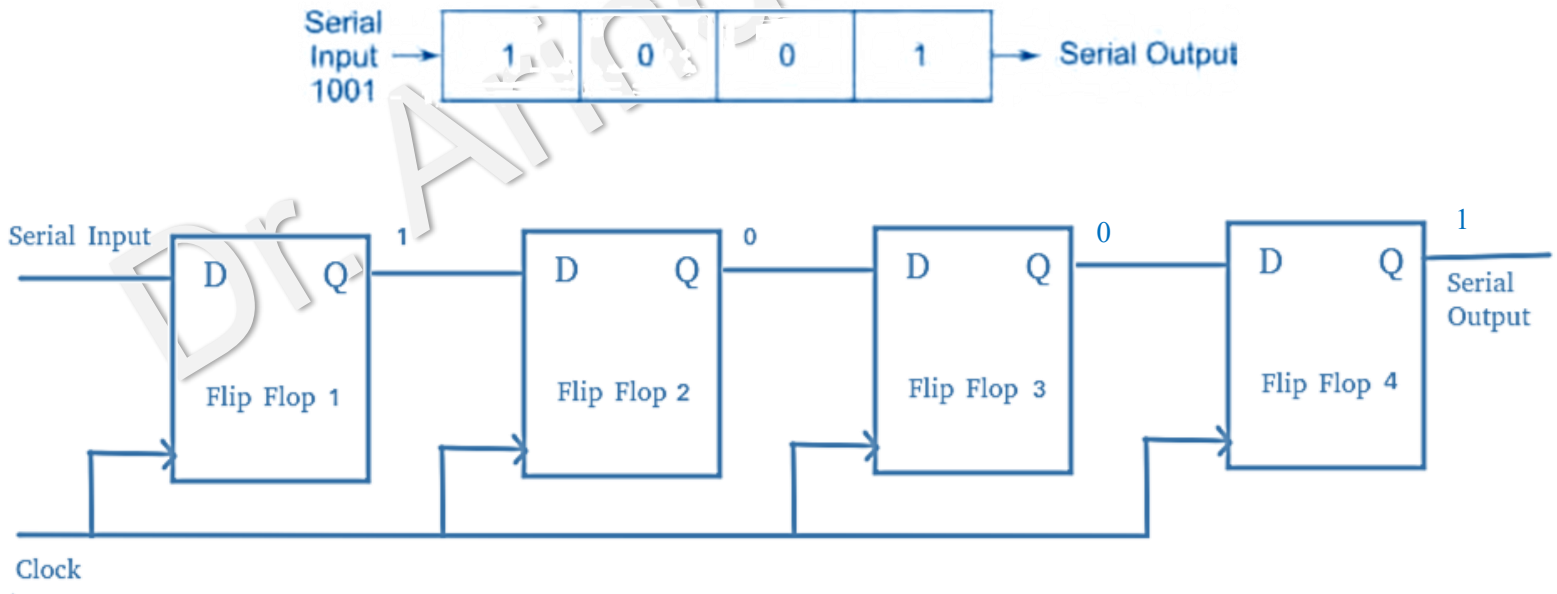


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Registers

Serial-In Serial-Out (SISO) Shift Register

- Data enters and leaves one bit at a time in a serial sequence.
- The transfer of new information into a register is referred to as loading or updating the register.
- Typically made of multiple D flip-flops connected in series and synchronised by a common clock.
- The main use of a SISO is to act as a delay element.

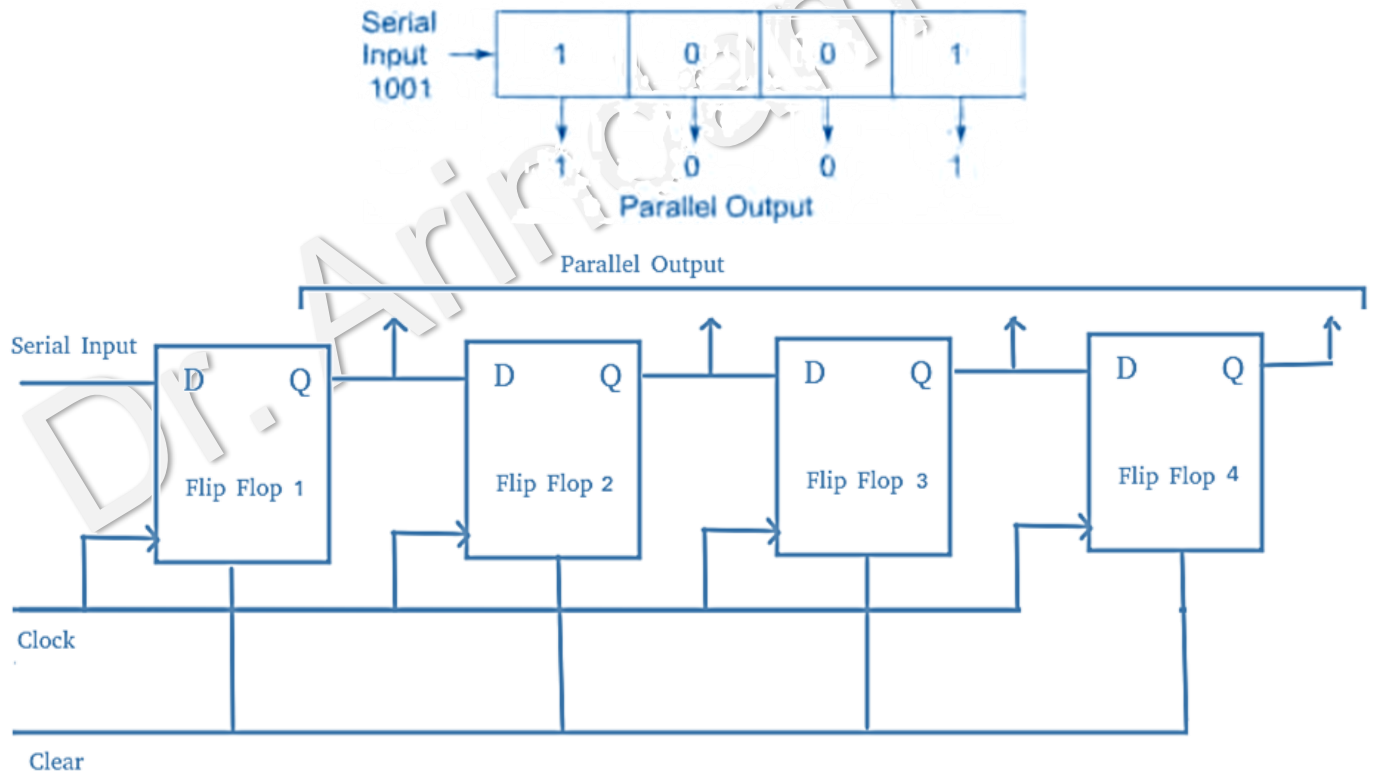


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Registers

Serial-In Parallel-Out (SIPO) Shift Register

- Data is input one bit at a time and outputs all stored bits in parallel.
- Consists of connected D flip-flops with a common clock and usually a clear (CLR) signal to reset all flip-flops.
- Mainly used to convert serial data into parallel data.

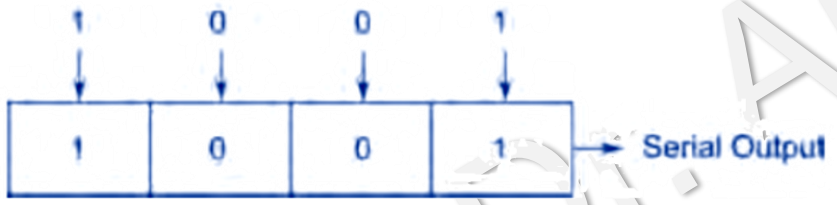


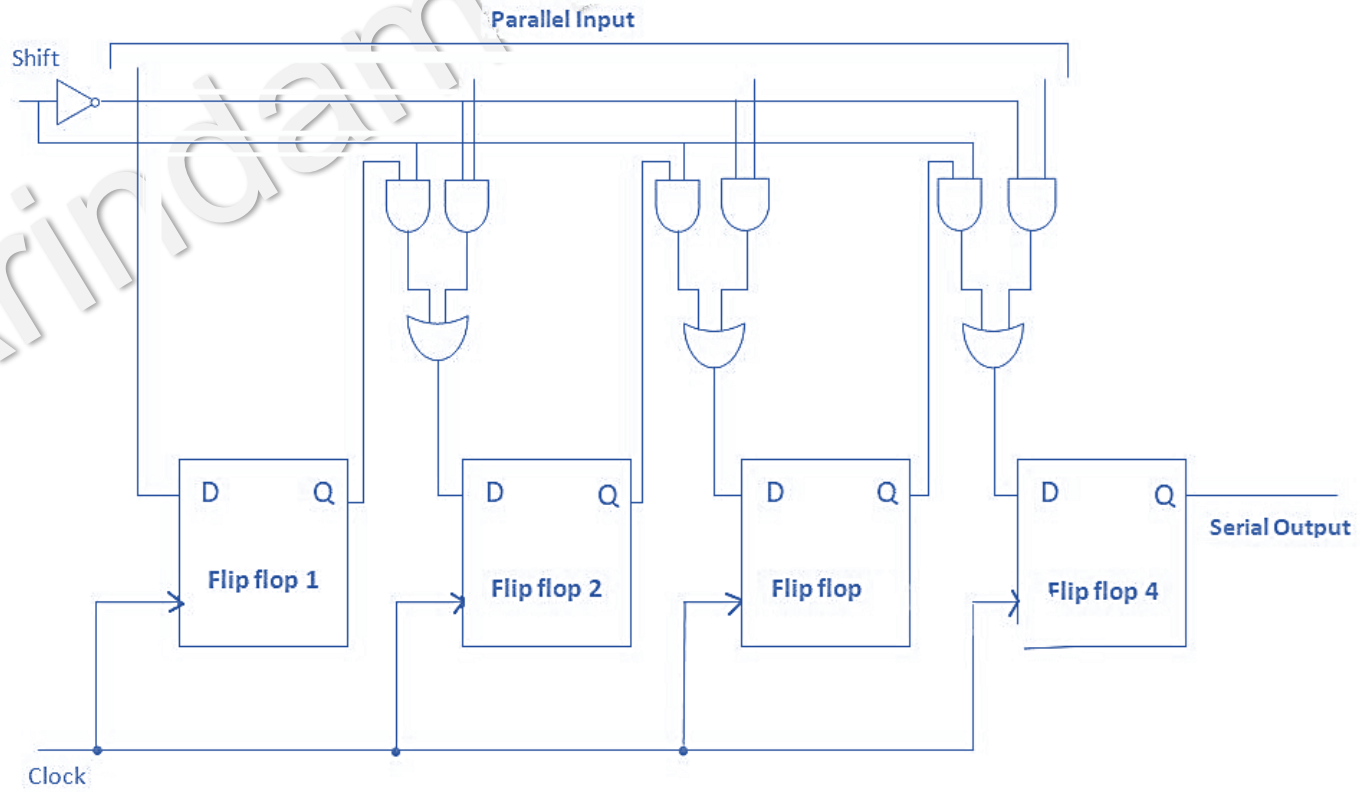
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Registers

Parallel-In Serial-Out (PISO) Shift Register

- Data is loaded in parallel and then shifted out one bit at a time serially.
- If all the bits of the register are loaded simultaneously with a common clock pulse, the loading is done in parallel.
- Consists of D flip-flops with inputs controlled by multiplexers selecting between parallel data and previous flip-flop output, all clocked together.



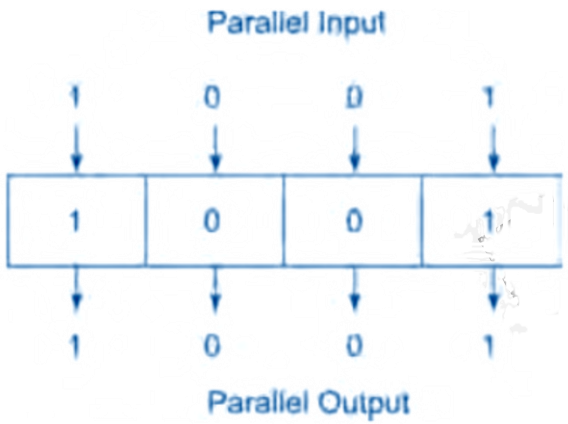


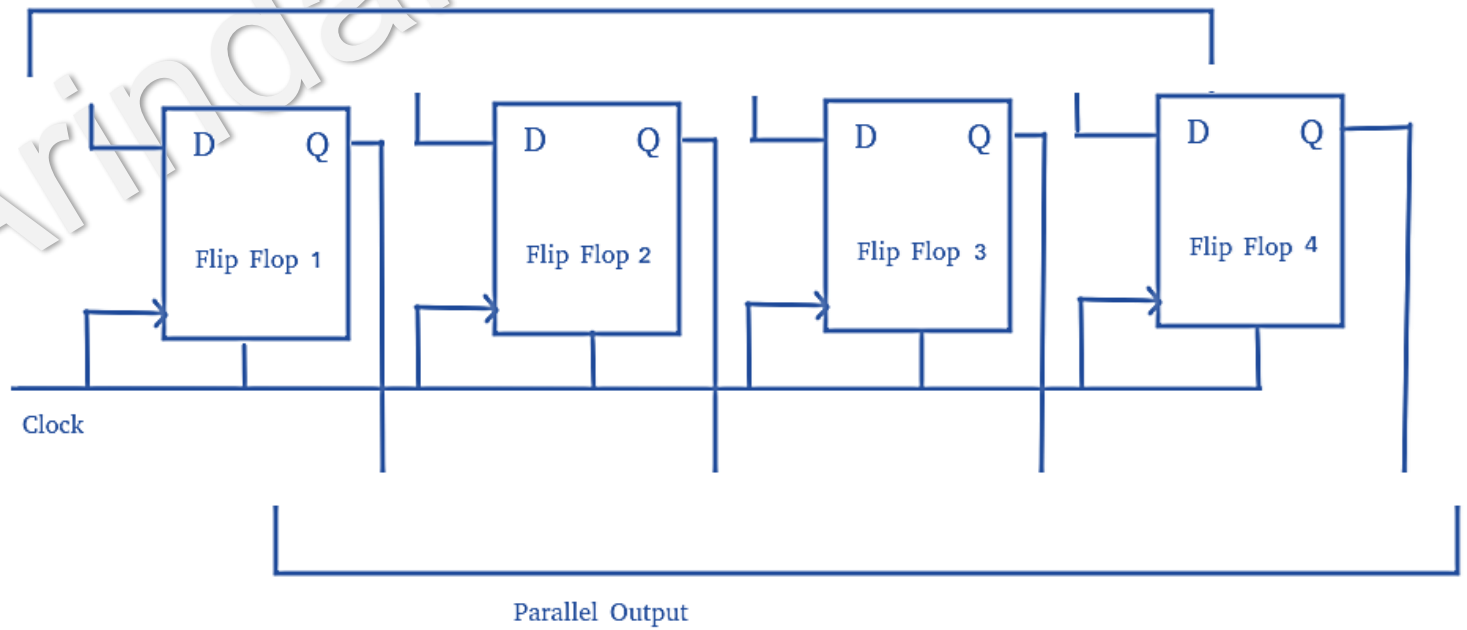
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Registers

Parallel-In Parallel-Out (PIPO) Shift Register

- Data is loaded and retrieved in parallel from each flip-flop at the same time.
- Includes D flip-flops with no interconnections, controlled by common clock and clear (CLR) signals.
- It is used as a temporary storage device and also acts as a delay element.





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Registers

Example: A 4-bit SISO shift register is initially loaded with $Q_3\ Q_2\ Q_1\ Q_0 = 1\ 1\ 0\ 1$. The serial input sequence applied to the register at successive positive clock edges is 0, 1, 1, 0, 1, 0. The register shifts from Q_3 towards Q_0, with the serial input entering Q_3. Determine the contents of the register after each clock pulse and find the value stored in the register after the 6th clock pulse.

Clock pulse	Serial input	Q_3	Q_2	Q_1	Q_0	Register contents
Initial	—	1	1	0	1	1101
1	0	0	1	1	0	0110
2	1	1	0	1	1	1011
3	1	1	1	0	1	1101
4	0	0	1	1	0	0110
5	1	1	0	1	1	1011
6	0	0	1	0	1	0101

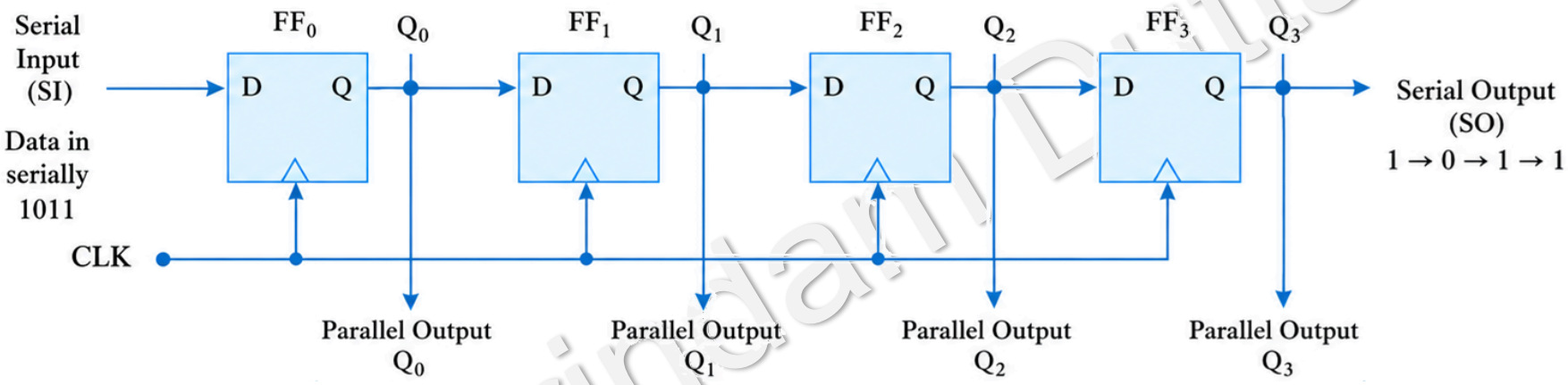
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So, after the 6th clock pulse = 0101

$$1101 \xrightarrow{0} 0110 \xrightarrow{1} 1011 \xrightarrow{1} 1101 \xrightarrow{0} 0110 \xrightarrow{1} 1011 \xrightarrow{0} \boxed{0101}$$

Registers

Example: **A 4-bit data word 1011 is to be transmitted serially, one bit at a time, on successive clock cycles. At the same time, the transmitted data must be available in parallel form after four clock cycles. Design a single digital system using shift registers that performs both operations. Assume a right shift register.**



Clock	Serial Input	Q_0	Q_1	Q_2	Q_3	Register contents
Initial	—	0	0	0	0	0000
1st	1	1	0	0	0	1000
2nd	1	1	1	0	0	1100
3rd	0	0	1	1	0	0110
4th	1	1	0	1	1	1011

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Counters

- A register that goes through a prescribed sequence of states upon the application of input pulses.
- The input pulses may be clock pulses, or they may originate from some external source and may occur at a fixed interval of time or at random.
- A counter that follows the binary number sequence is called a binary counter.
- An *n*-bit binary counter consists of *n* flip-flops and can count in binary from 0 through 2^n - 1.
- Available in two categories: ripple or asynchronous counters and synchronous counters.
- In a ripple counter, a flip-flop output transition serves as a source for triggering other flip-flops. In other words, the *C* input of some or all flip-flops are triggered, not by the common clock pulses, but rather by the transition that occurs in other flip-flop outputs.
- In a synchronous counter, the *C* inputs of all flip-flops receive the common clock.
- Can count in increasing order (up counter) and/or in decreasing order (down counter).

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Ripple Counters

Binary Ripple Counters

- Consists of a series connection of complementing flip-flops, with the output of each flip-flop connected to the *C* input of the next higher order flip-flop.
- The flip-flop holding the least significant bit receives the incoming count pulses.
- A complementing flip-flop can be obtained from a JK flip-flop with the J and K inputs tied together or from a T flip-flop.
- A third possibility is to use a D flip-flop with the complement output connected to the *D* input. In this way, the *D* input is always the complement of the present state, and the next clock pulse will cause the flip-flop to complement.
- In a 4-bit binary counter, the count starts with binary 0 and increments by 1 with each count pulse input. After the count of 15, the counter goes back to 0 to repeat the count (Table and circuits in next slide).

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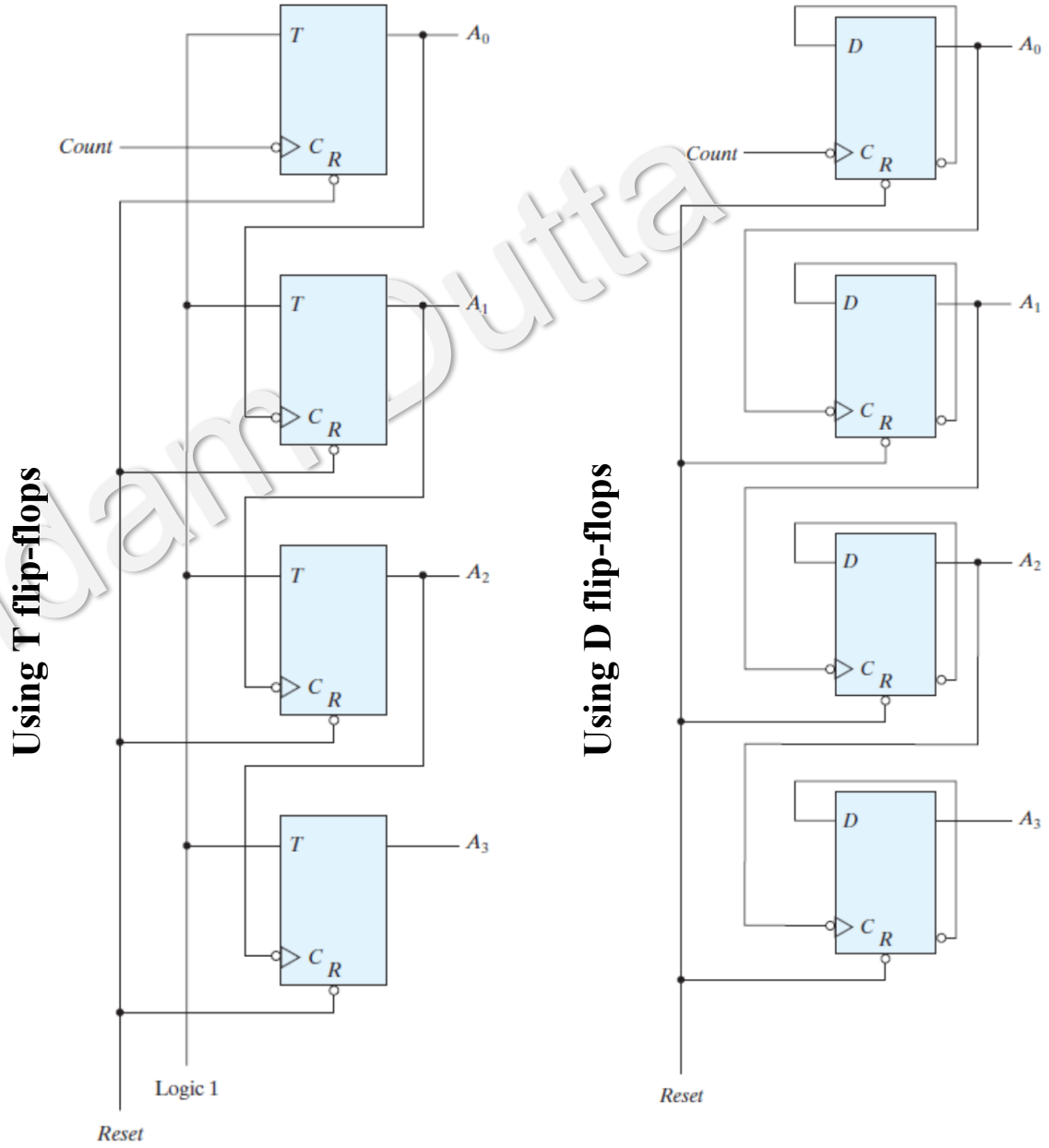
Ripple Counters

Binary Ripple Counters

Binary Count Sequence

A_3	A_2	A_1	A_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0

The bubble next to *C* indicates that the flip-flops respond to the negative-edge transition of the input, which occurs when the output of the previous flip-flop to which *C* is connected goes from 1 to 0.



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Ripple Counters

Modulus Counters

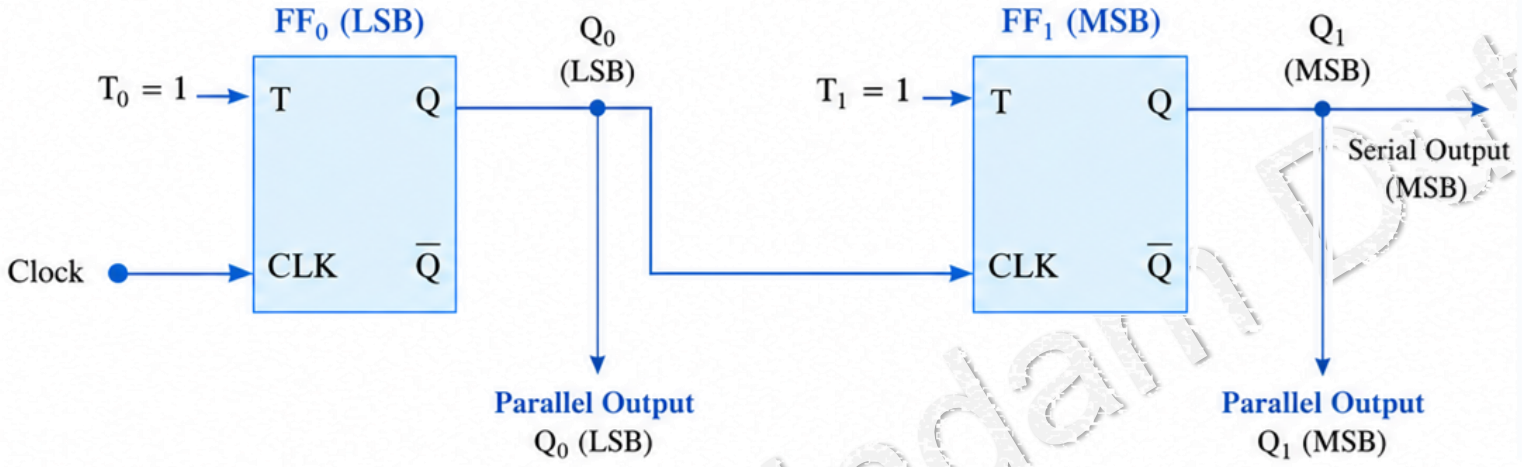
- Counts through a specific number of n states (0 to n-1) before resetting itself back to zero, or some known starting point.
- The number of states in a counter is known as its modulo number or mod number. Accordingly, a 2-bit counter is a Mod-4 counter, a 3-bit counter is a Mod-8 counter and so on.
- Thus, the 4-bit binary ripple counter explained in the previous slides is a Mod-16 counter.
- A Mod-2 counter counts 2 states (0 and 1, then returns to 0), a Mod-4 counter counts 4 states (00, 01, 10, 11 and returns to 00), a Mod-8 counter counts 8 states (000 – 111 and returns to 000), etc.
- A counter with m flip-flops can have 2^m states.
- If number of states (n) is not a power of 2, then we will need m flip-flops, where 2^m is the smallest power of 2 greater than or equal to n.
- Widely used to create digital circuits in frequency division and counting applications.

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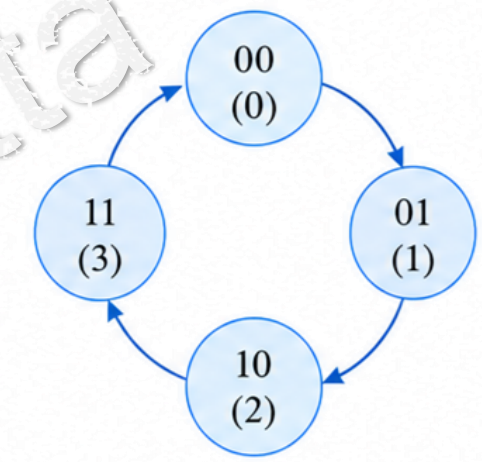
Ripple Counters

Mod-4 Ripple Counter

Circuit Diagram



State Diagram



Operation (Counting Sequence)

Clock Pulse	T_1 (MSB)	T_0 (LSB)	Present State		Next State		Count (Decimal)	Counter Sequence (Q_1 Q_0)
			Q_1 (MSB)	Q_0 (LSB)	Q_1 (MSB)	Q_0 (LSB)		
Initial	1	1	0	0	0	1	0	00
1st Pulse	1	1	0	1	1	0	1	01
2nd Pulse	1	1	1	0	1	1	2	10
3rd Pulse	1	1	1	1	0	0	3	11
4th Pulse	1	1	0	0	0	1	0	00
…	…	…	…	…	…	…	…	…

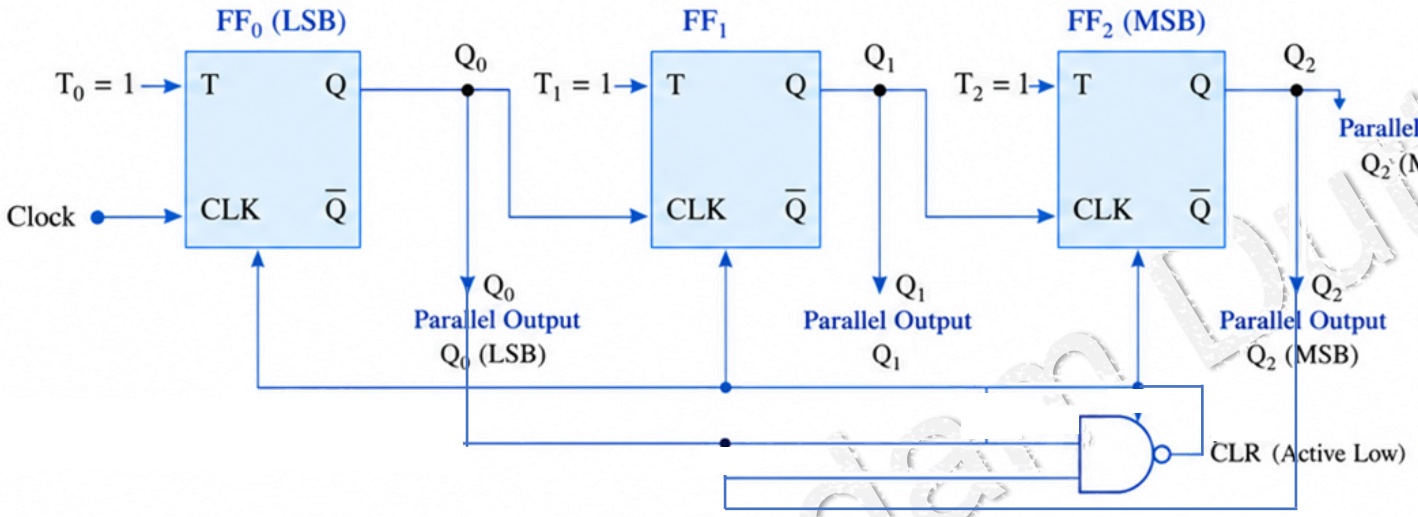
Notes:

- T_0 (LSB) toggles at every clock pulse.
- T_1 (MSB) toggles when Q_0 changes from 1 to 0.
- The counter sequence is $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0 \rightarrow \ldots$ (MOD-4 counter).

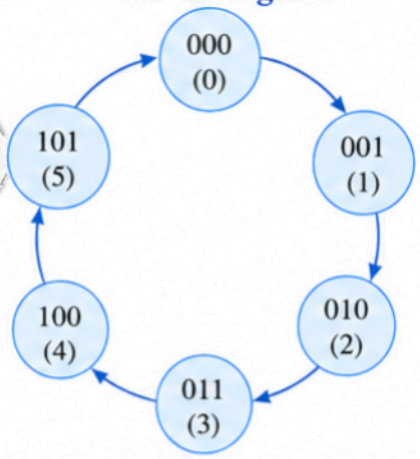
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Ripple Counters

Mod-6 Ripple Counter



State Diagram



Reset Condition

When the counter reaches state 101 (decimal 5),
$Q_2 = 1$ and $Q_0 = 1$
⇒ NAND output = 0
⇒ All flip-flops are cleared asynchronously to 000 (0).

Operation (Counting Sequence)

Clock Pulse	Present State			Next State			Count (Decimal)	Counter Sequence ($Q_2\ Q_1\ Q_0$)
	Q_2 (MSB)	Q_1	Q_0 (LSB)	Q_2 (MSB)	Q_1	Q_0 (LSB)		
Initial	0	0	0	0	0	1	0	000
1st Pulse	0	0	1	0	1	0	1	001
2nd Pulse	0	1	0	0	1	1	2	010
3rd Pulse	0	1	1	1	0	0	3	011
4th Pulse	1	0	0	1	0	1	4	100
5th Pulse	1	0	1	0	0	0	5	101 → 000 (reset)
6th Pulse	0	0	0	0	0	1	0	000
…	…	…	…	…	…	…	…	…

Notes:

- T_0, T_1, T_2 are tied to 1, so each flip-flop toggles when its clock input receives a falling (1 – 0) edge from the previous stage.
- The counter normally counts 0 to 7. State 5 (101) is detected by Q_2 and Q_0 both being 1.
- The asynchronous clear brings the counter back to 000.
- Therefore the sequence is: 0 → 1 → 2 → 3 → 4 → 5 → 0 → … (MOD-6 counter).

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Module 4 – Lecture 2

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Synchronous Counters

- Synchronous counters are different from ripple counters in that clock pulses are applied to the inputs of all flip-flops.
- A common clock triggers all flip-flops simultaneously, rather than one at a time in succession as in a ripple counter.
- Can be up counter, down counter, or up/down counter.
- The decision whether a flip-flop is to be complemented is determined from the values of the data inputs, such as T or J and K at the time of the clock edge.
- If $T = 0$ or $J = K = 0$, the flip-flop does not change state. If $T = 1$ or $J = K = 1$, the flip-flop complements.
- The design procedure for synchronous counters in the form of a 3-bit up binary counter has been explained in Module 3.

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Synchronous Counters

Up Binary Counter

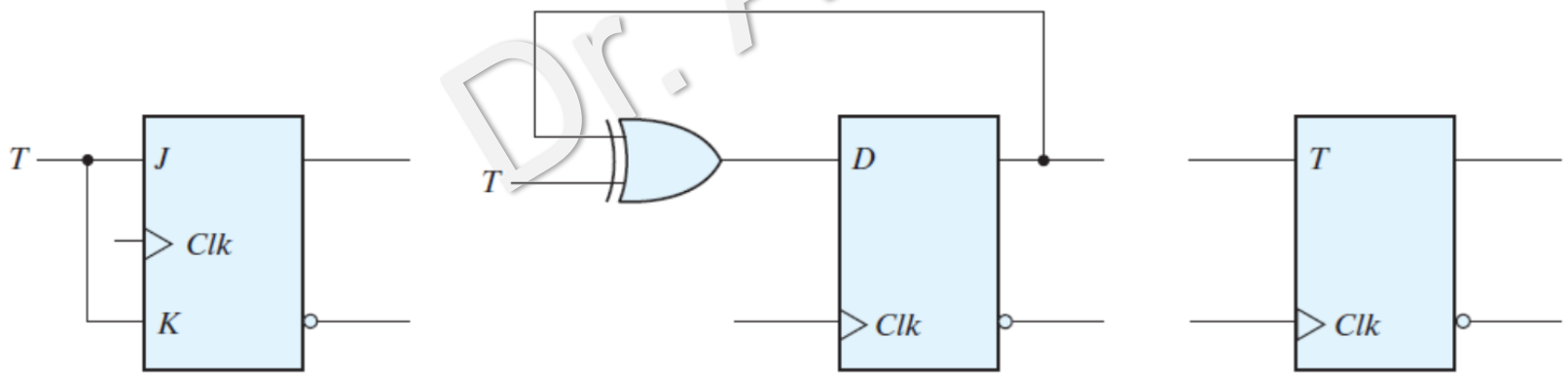
- In a synchronous binary counter, the flip-flop in the least significant position is complemented with every pulse.
- *A flip-flop in any other position is complemented when all the bits in the lower significant positions are equal to 1*. For example, if the present state of a four-bit (Mod-16) up counter is $A_3A_2A_1A_0 = 0011$, the next count is 0100. A_0 is always complemented. A_1 is complemented because the present state of $A_0 = 1$. A_2 is complemented because the present state of $A_1A_0 = 11$. However, A_3 is not complemented, because the present state of $A_2A_1A_0 = 011$, which does not give an all-1's condition.
- Synchronous binary counters have a regular pattern and can be constructed with complementing flip-flops and gates.
- From the circuit in next slide, the *C* inputs of all flip-flops are connected to a common clock. The counter is enabled by *Count_enable*. If the enable input is 0, all *J* and *K* inputs are equal to 0 and the clock does not change the state of the counter. The first stage, A_0, has its *J* and *K* equal to 1 if the counter is enabled. The other *J* and *K* inputs are equal to 1 if all previous least significant stages are equal to 1 and the count is enabled.
- The chain of AND gates generates the required logic for the *J* and *K* inputs in each stage. The counter can be extended to any number of stages, with each stage having an additional flip-flop and an AND gate that gives an output of 1 if all previous flip-flop outputs are 1.

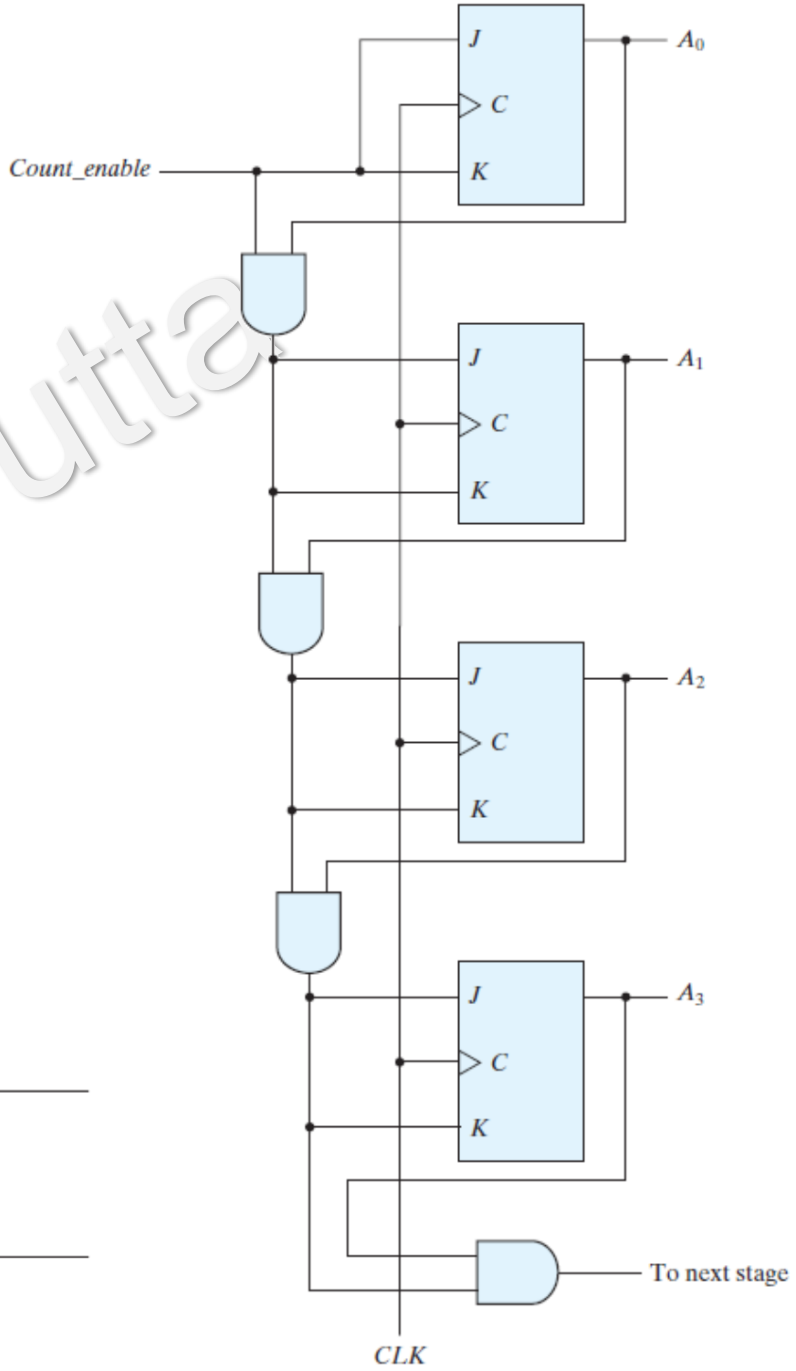
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Synchronous Counters

Up Binary Counter

- The polarity of the clock is not essential here, but it is with the ripple counter.
- The synchronous counter can be triggered with either the positive or the negative clock edge.
- In this example, the flip-flops trigger on the positive edge of the clock.
- The complementing flip-flops in a binary counter can be of either the *JK* type, the *T* type, or the *D* type with XOR gates. The equivalency of the three types is indicated below.





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Synchronous Counters

Down Binary Counter

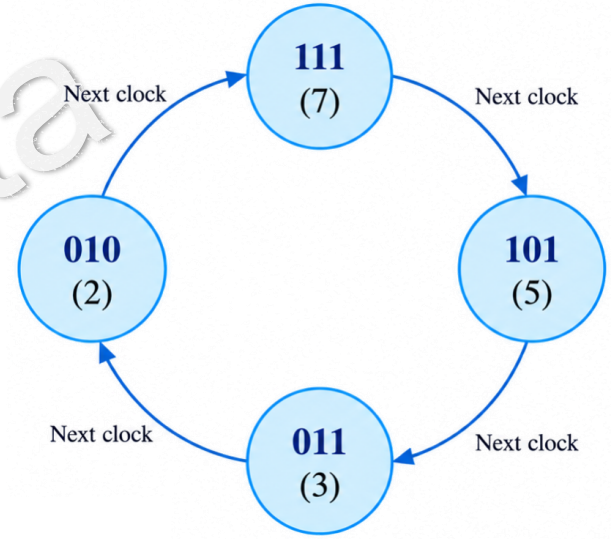
- A synchronous down (or countdown) binary counter goes through the binary states in reverse order, for example, from 1111 down to 0000 and back to 1111 to repeat the count in case of a 4-bit counter.
- It is possible to design a down counter in the usual manner, but the result is predictable by inspection of the downward binary count.
- The bit in the least significant position is complemented with each pulse. A bit in any other position is complemented if all lower significant bits are equal to 0.
- For example, the next state after the present state of 0100 is 0011. The least significant bit is always complemented. The second significant bit is complemented because the first bit is 0. The third significant bit is complemented because the first two bits are equal to 0. But the fourth bit does not change, because not all lower significant bits are equal to 0.
- A down binary counter can be constructed as shown in the previous slide, except that the inputs to the AND gates must come from the complemented outputs, instead of the normal outputs, of the previous flip-flops.

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Synchronous Counters

Example: Design a synchronous down counter using T flipflops which counts the prime numbers between 10 and 1.

- There are 4 prime numbers in the given range: 7, 5, 3, 2 (four states).
- Since 7 is the highest number, a 3-bit system would be sufficient (as 7 = 111).
- Count sequence: 7 (111) – 5 (101) – 3 (011) – 2 (010) – 7 (111) – ….
- Number of flipflops required = number of bits = 3.



State Diagram

State Table

Clk	Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	T_C	T_B	T_A
Initial	1	1	1	1	0	1	0	1	0
1	1	0	1	0	1	1	1	1	0
2	0	1	1	0	1	0	0	0	1
3	0	1	0	1	1	1	1	0	1

Q_c \ Q_BQ_A	00	01	11	10
0	x	x	0	1
1	x	1	0	x

$T_C = Q_A' + Q_B'$

Q_c \ Q_BQ_A	00	01	11	10
0	x	x	0	0
1	x	1	1	x

$T_B = Q_C$

Q_c \ Q_BQ_A	00	01	11	10
0	x	x	1	1
1	x	0	0	x

$T_A = Q_C'$

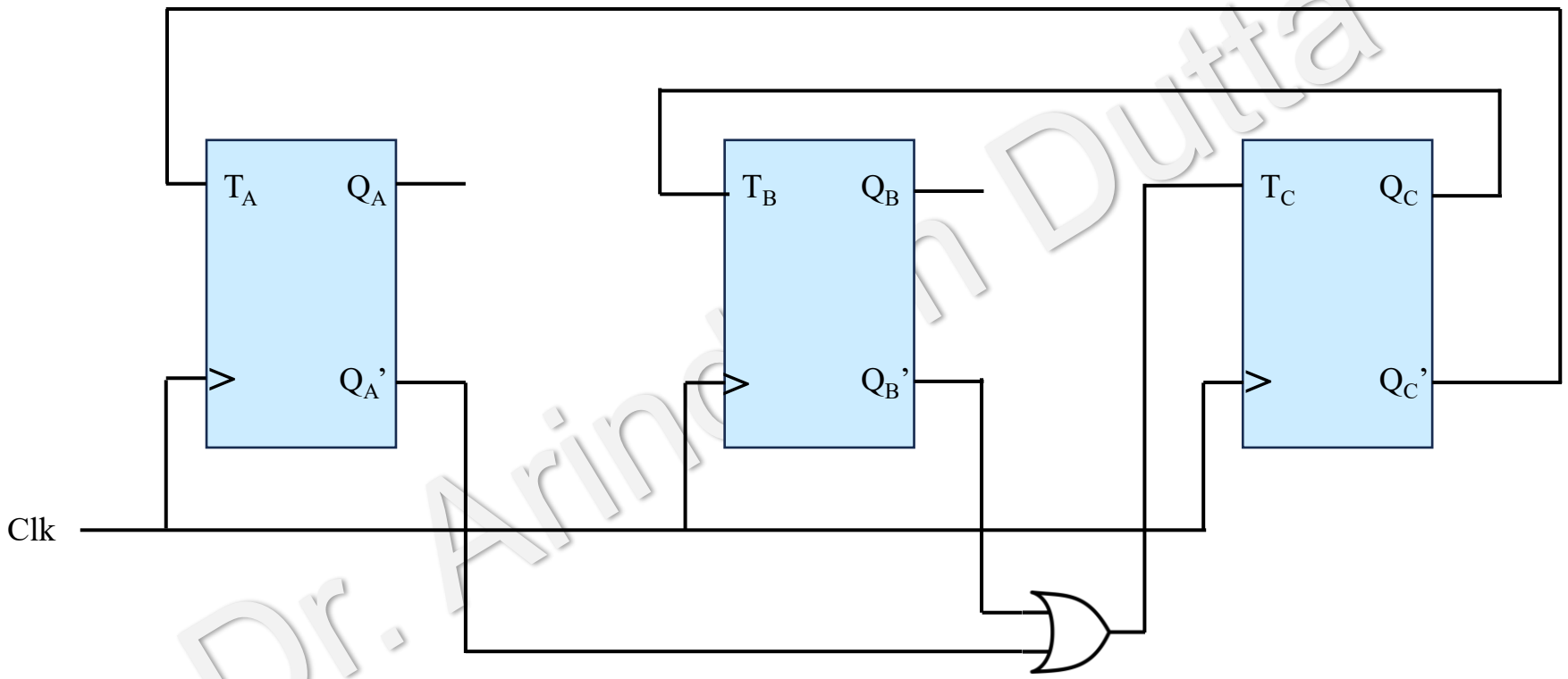
Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table

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Synchronous Counters

Final Circuit:



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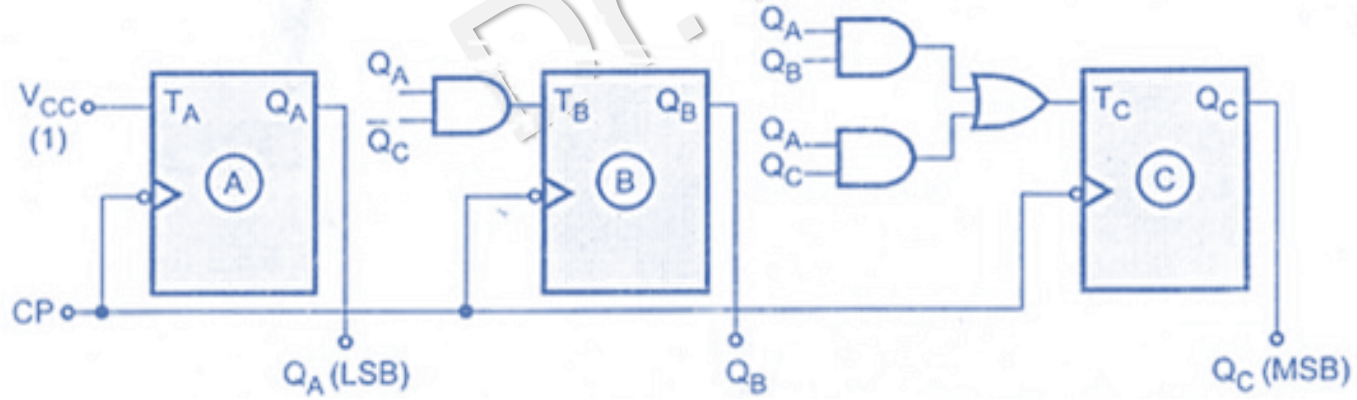
Synchronous Counters

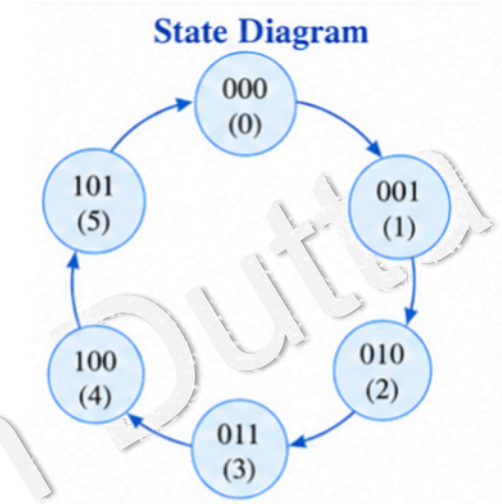
Mod-6 Up Counter

- States (N) = 6. So, number of flip-flops = 3 (since, $2^3 = 8$)
- States that are not used are treated as don't-care conditions
- Type of flip-flop: T

State Table

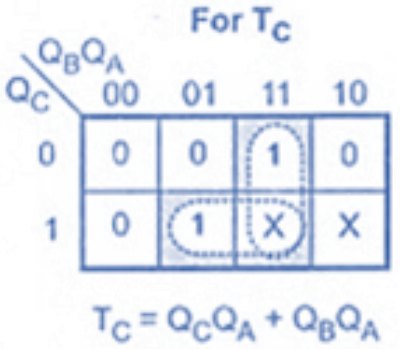
Clk	Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	T_C	T_B	T_A
Initial	0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	0	1	1
2	0	1	0	0	1	1	0	0	1
3	0	1	1	1	0	0	1	1	1
4	1	0	0	1	0	1	0	0	1
5	1	0	1	0	0	0	1	0	1

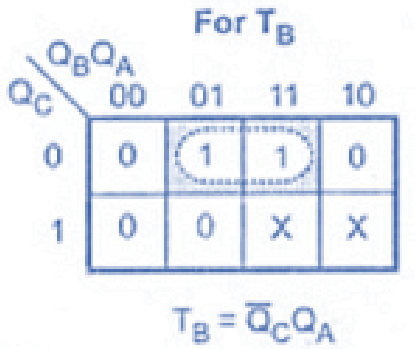


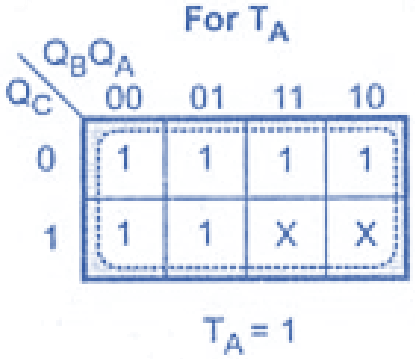


Excitation Table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0





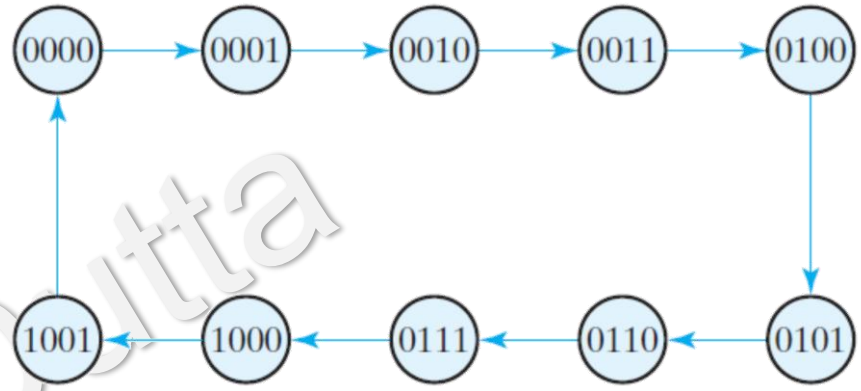


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Synchronous Counters

Mod-10 (or BCD or Decade) Up Counter

- Counts in binary-coded decimal from 0000 to 1001 and back to 0000
- States (N) = 10. So, number of flip-flops = 4 (since, $2^4 = 16$)
- Type of flip-flop: T



State Diagram

State Table

Present State				Next State				Flip-Flop Inputs			
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table

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Synchronous Counters

Mod-10 (or BCD or Decade) Up Counter

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	X	X	X	X
10	0	1	X	X

$T_3 = Q_3Q_0 + Q_2Q_1Q_0$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	X	X	X	X
10	0	0	X	X

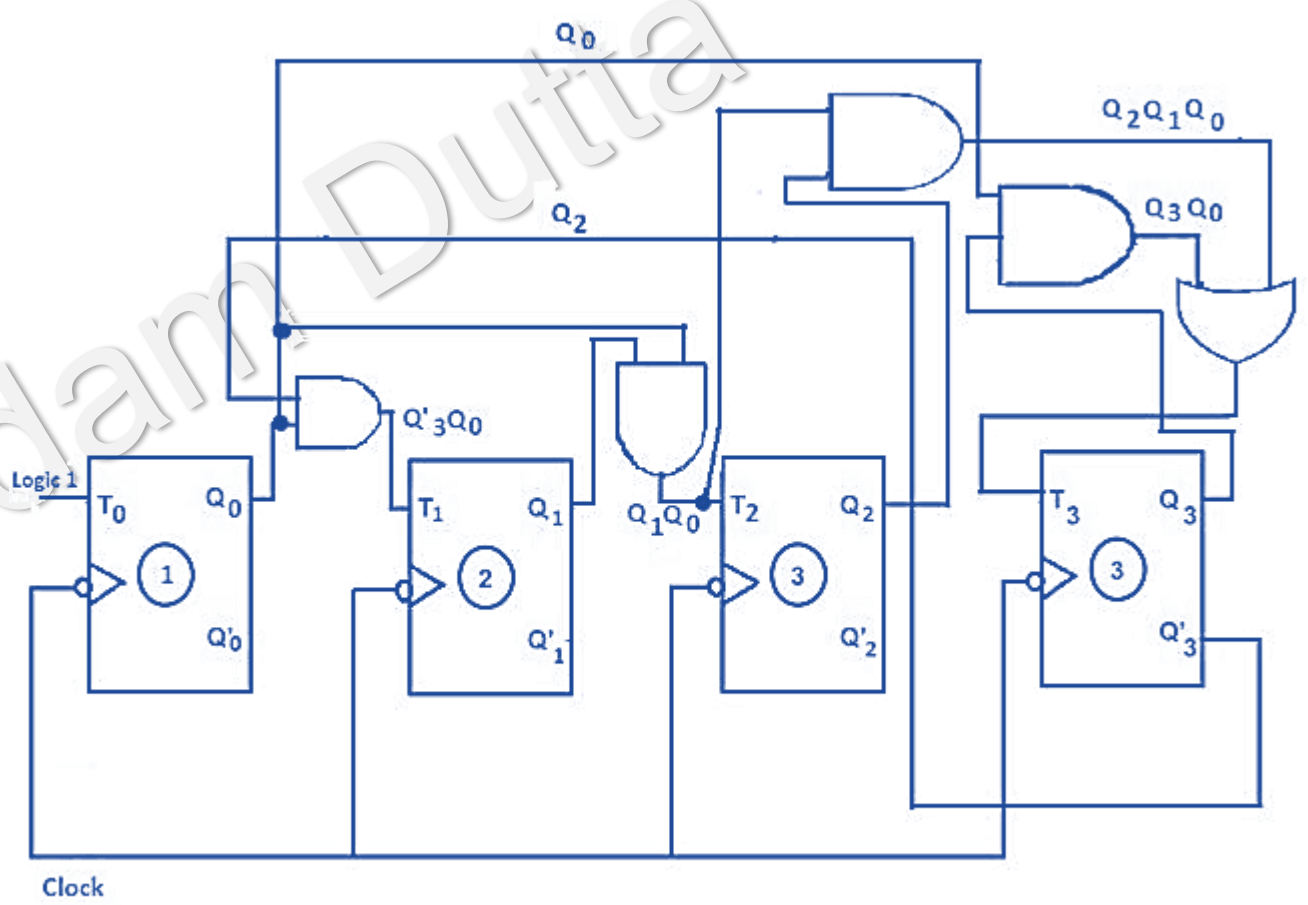
$T_2 = Q_1Q_0$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	X	X	X	X
10	0	0	X	X

$T_1 = Q'_3Q_0$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$T_0 = 1$



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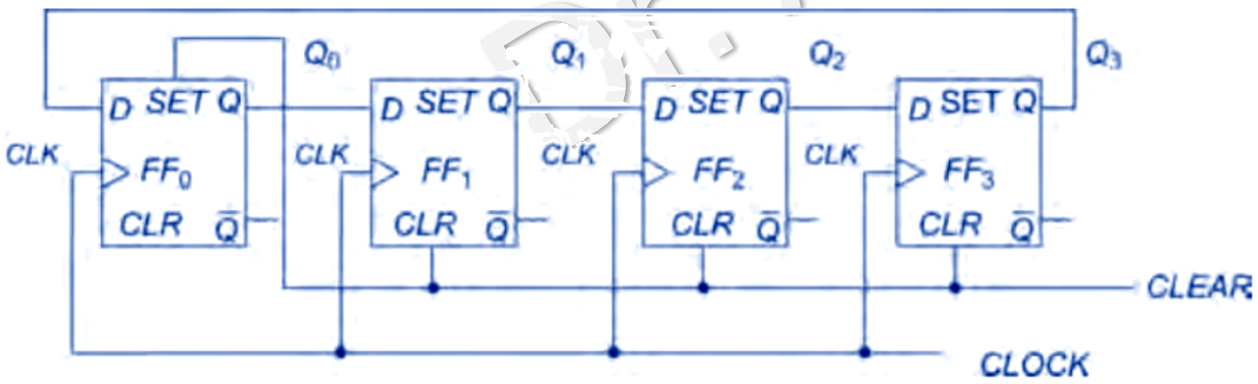
Synchronous Counters

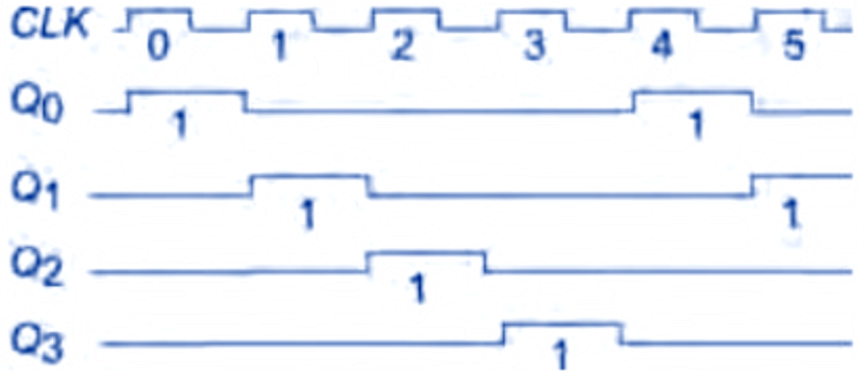
Ring Counter

- It is a circulating shift register in which the output of the most significant stage is fed back to the input of the least significant stage.

Clock	Q_0	Q_1	Q_2	Q_3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1

- For example, the figure shows a 4-bit ring counter constructed using D flip-flops.
- The single-bit output of each stage is shifted into the next stage, say, on the positive edge of a clock pulse.
- Initially, FF_0 is set and all other flip-flops are reset. Therefore, Q_0 is 1 and Q_1, Q_2, and Q_3 are 0. After application of clock pulse, output Q_0 will be shifted to Q_1 and Q_1 will be 1. Similarly, just after the clock pulse-2, output Q_2 will be 1 and after third clock pulse Q_3 will be 1.
- As the 4-bit ring counter utilizes only 4 different states, it can be considered as a mod-4 counter.





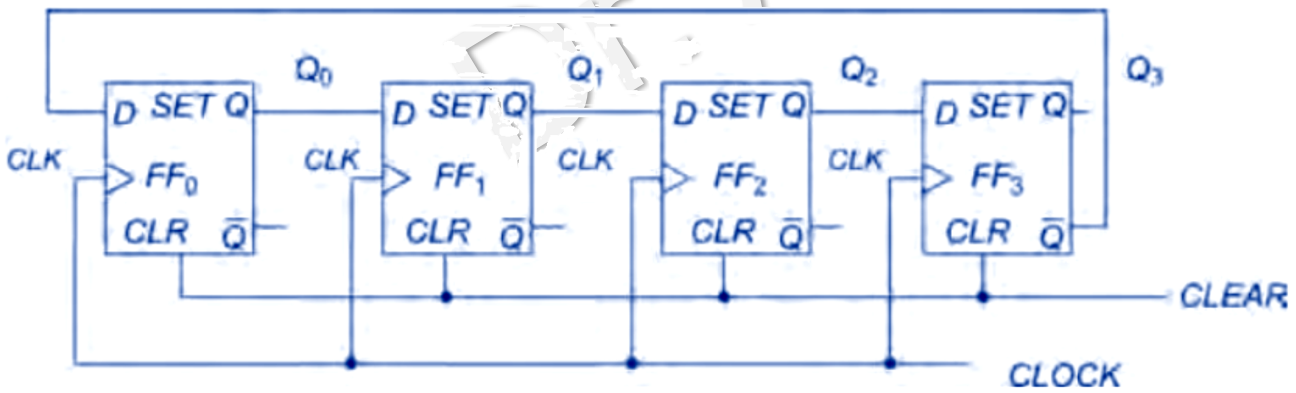
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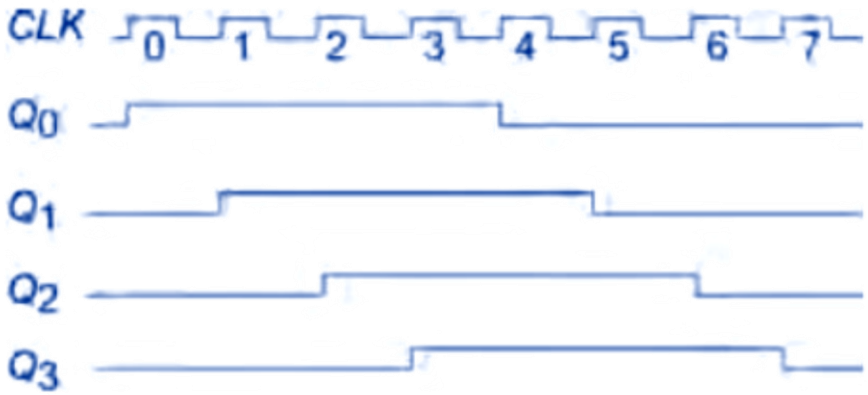
Synchronous Counters

Johnson Counter

- Here, inverted output of the last stage flip-flop is fed back to the input of the first stage flip-flop.
- A four-bit Johnson counter has eight counting sequence (states) and five-bit Johnson counter has ten counting sequence.
- Therefore '*n*' bit Johnson counter should have a count sequence of length '2*n*' and it may be called as mod-2*n* counter.
- The number of flip-flops needed is one-half the number of timing signals.
- These counters are also known as twisted ring counters or switch-tail ring counter.

Clock	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1





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DIGITAL LOGIC DESIGN

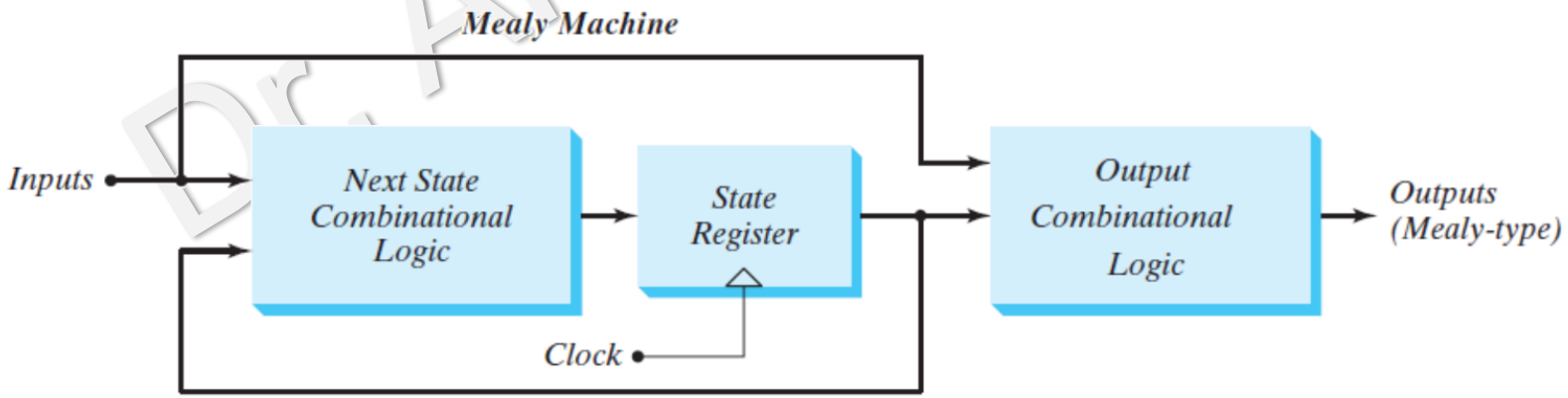
Module 4 – Lecture 3

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Mealy and Moore Machines

- The most general model of a sequential circuit has inputs, outputs, and internal states.
- There are two models of sequential circuits: the **Mealy model** and the **Moore model**. They differ only in the way the output is generated. A circuit may have both types of outputs.
- The two models of a sequential circuit are commonly referred to as a finite state machine, abbreviated FSM.
- In the Mealy model, the output is a function of both the present state and the input. The output of the Mealy machine is the value that is present immediately before the active edge of the clock.
- The Mealy model of a sequential circuit is referred to as a Mealy FSM or Mealy machine.

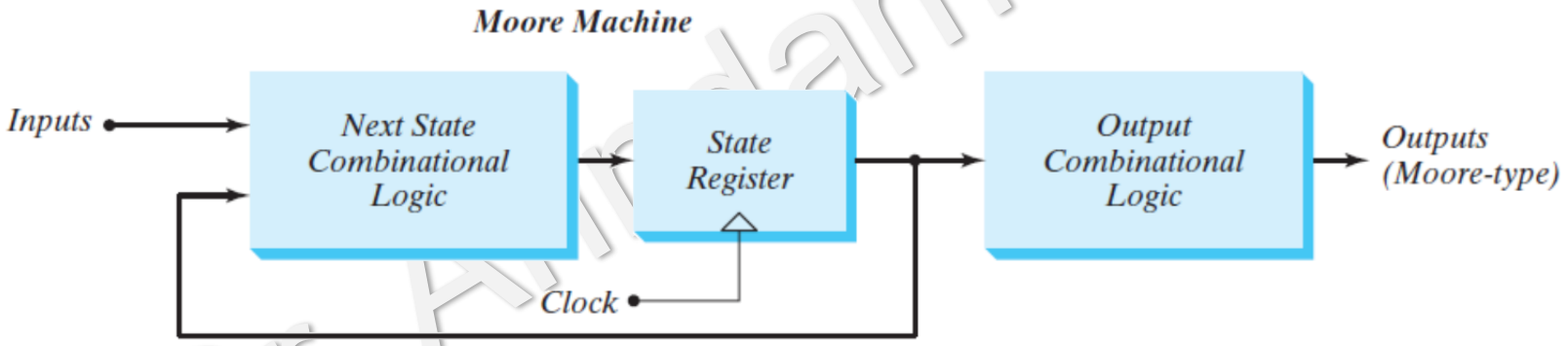


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Mealy and Moore Machines

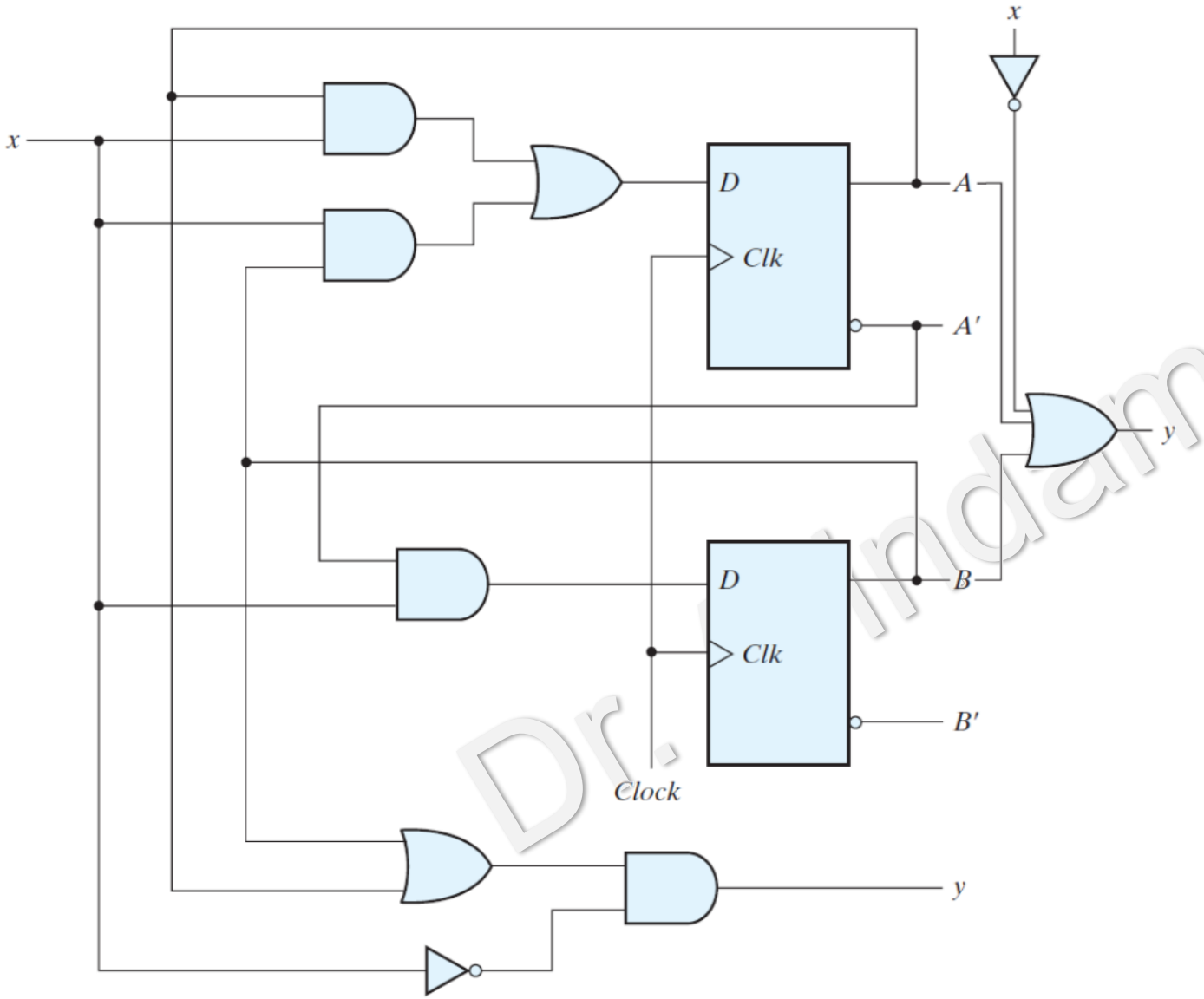
- In the Moore model, the output is a function of only the present state. The outputs of the sequential circuit are synchronized with the clock, because they depend only on flip-flop outputs that are synchronized with the clock.

- The Moore model is referred to as a Moore FSM or Moore machine.



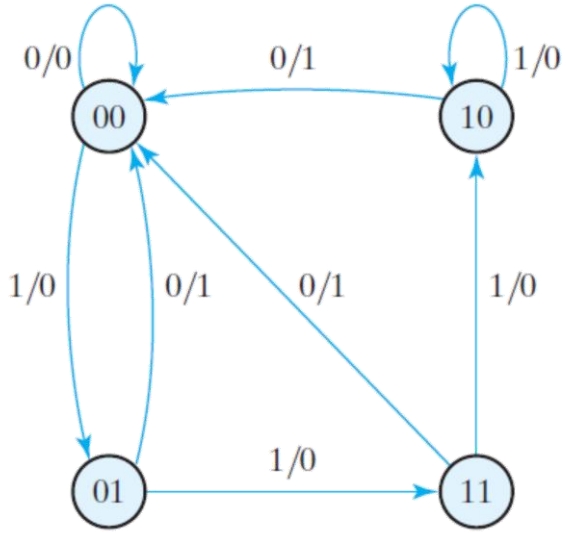
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Mealy Machine



Output y is a function of both input x and the present state of A and B

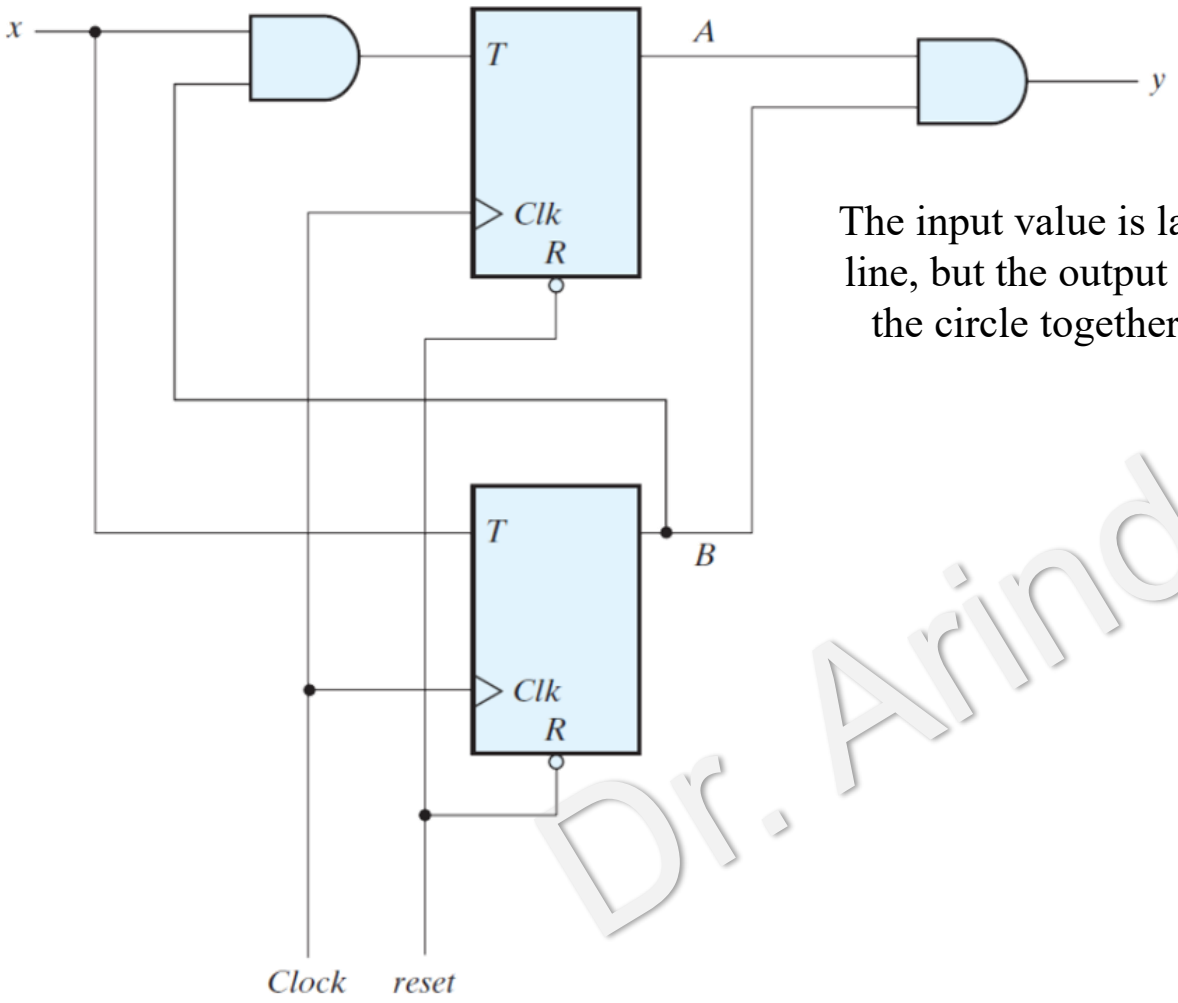
Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0



The input and output values are separated by a slash along the directed lines between the states.

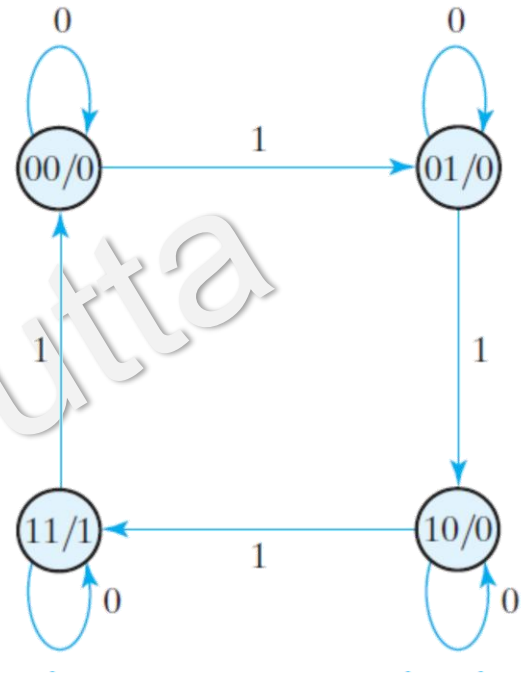
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Moore Machine



The output depends only on flip-flop values, and that makes it a function of the present state only.

The input value is labeled along the directed line, but the output value is indicated inside the circle together with the present state.

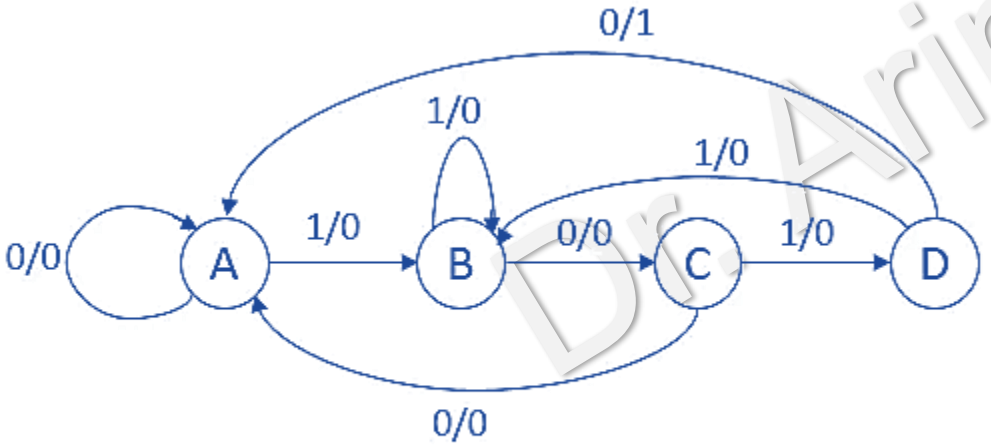


Present State		Input	Next State		Output
A	B	x	A	B	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	1

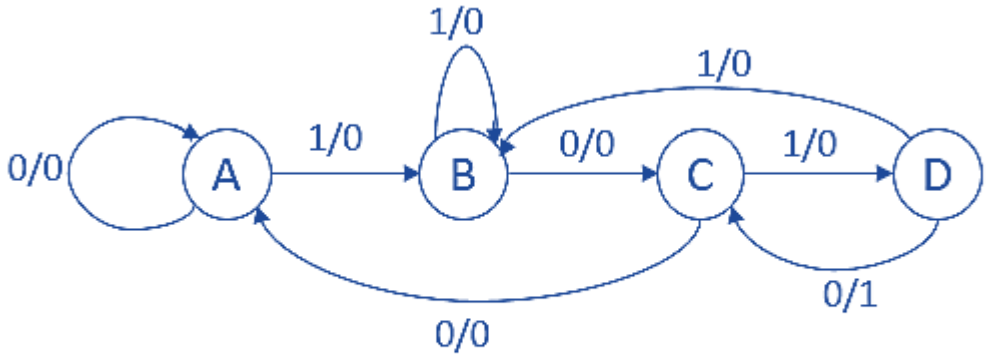
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Sequence Detector

- Digital circuit that detects some input signal sequences from a set of the binary data.
- One can determine whether incoming bits are equal to a prestored sequence, thus widely used in communication systems, data processing, and digital signal processing.
- Sequence detector is of two types: **Overlapping** (Mealy or Moore) and **Non-Overlapping** (Mealy or Moore)
- In an overlapping sequence detector, the last bits of one sequence can become the first bits of the next sequence.
- In a non-overlapping sequence detector, the last bits of one sequence do not become the first bits of the next sequence.



1010 Non-Overlapping Mealy Sequence Detector

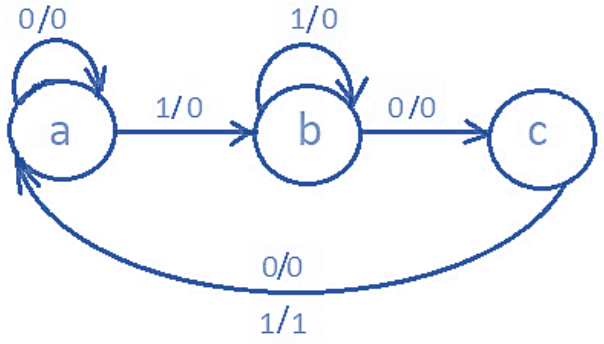


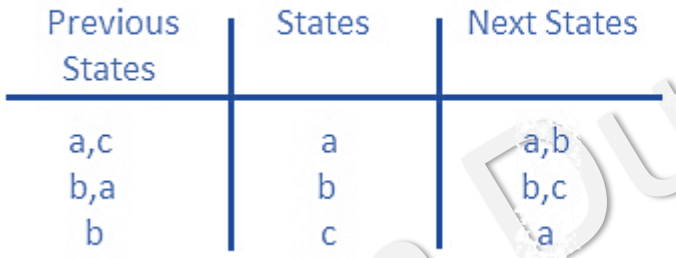
1010 Overlapping Mealy Sequence Detector

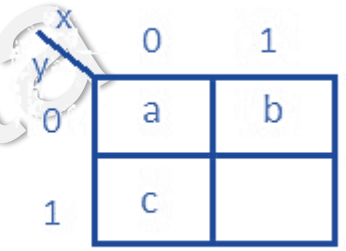
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Sequence Detector

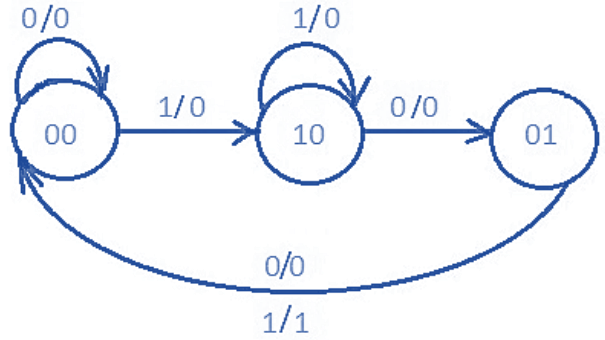
Example: Design a non-overlapping sequence detector (Mealy machine) that detects the string of bits 101.







Present States		i/p	Next States		Flip Flop Excitations		O/P
X	Y		X'	Y'	Dx	Dy	
0	0	0	0	0	0	0	0
0	0	1	1	0	1	0	0
0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	1
1	0	0	0	1	0	1	0
1	0	1	1	0	1	0	0
1	1	0	X	X	X	X	X
1	1	1	X	X	X	X	X



The state assignment has been done so as to keep adjacent assignments for the states having the same next states for a given input condition.

Alternatively, the problem can also be done with binary assignment (a = 00, b = 01, c = 10).

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Sequence Detector

…contd.

I \ XY	00	01	11	10
0	0	0	X	0
1	1	0	X	1

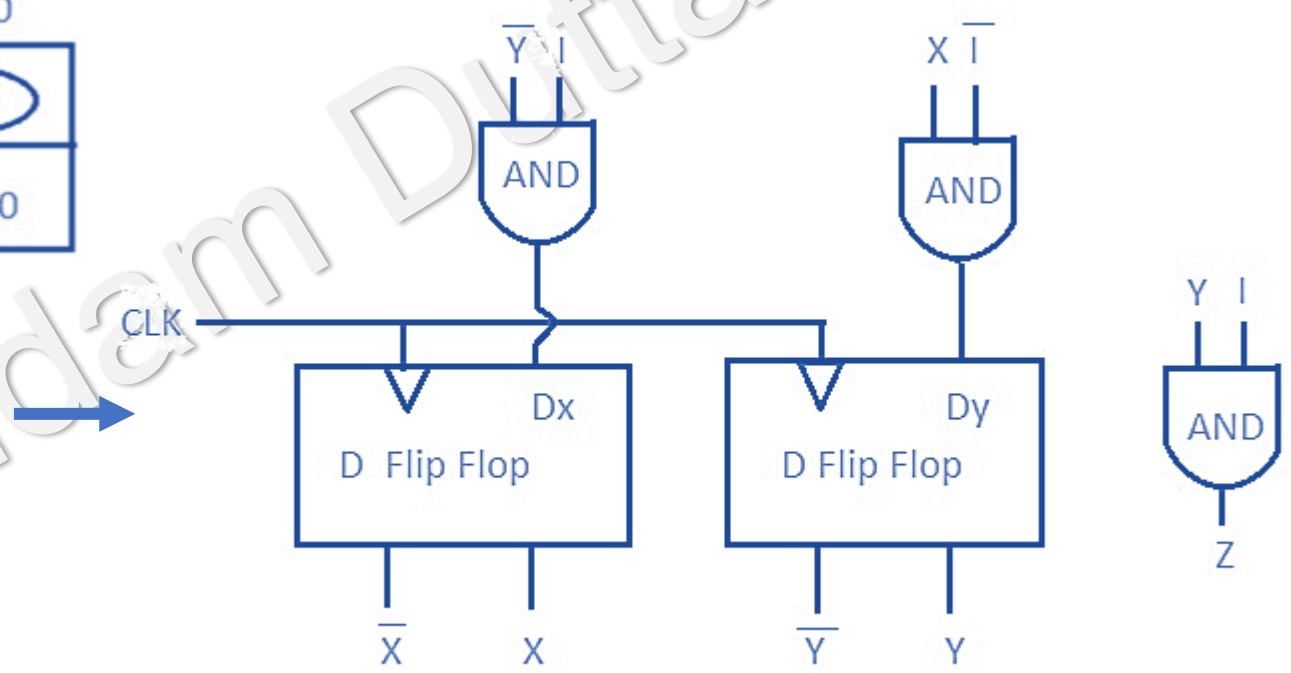
$D_x = \bar{Y}.I$

I \ XY	00	01	11	10
0	0	0	X	1
1	0	0	X	0

$D_y = X.\bar{I}$

I \ XY	00	01	11	10
0	0	0	X	0
1	0	1	X	0

$Z = Y.I$

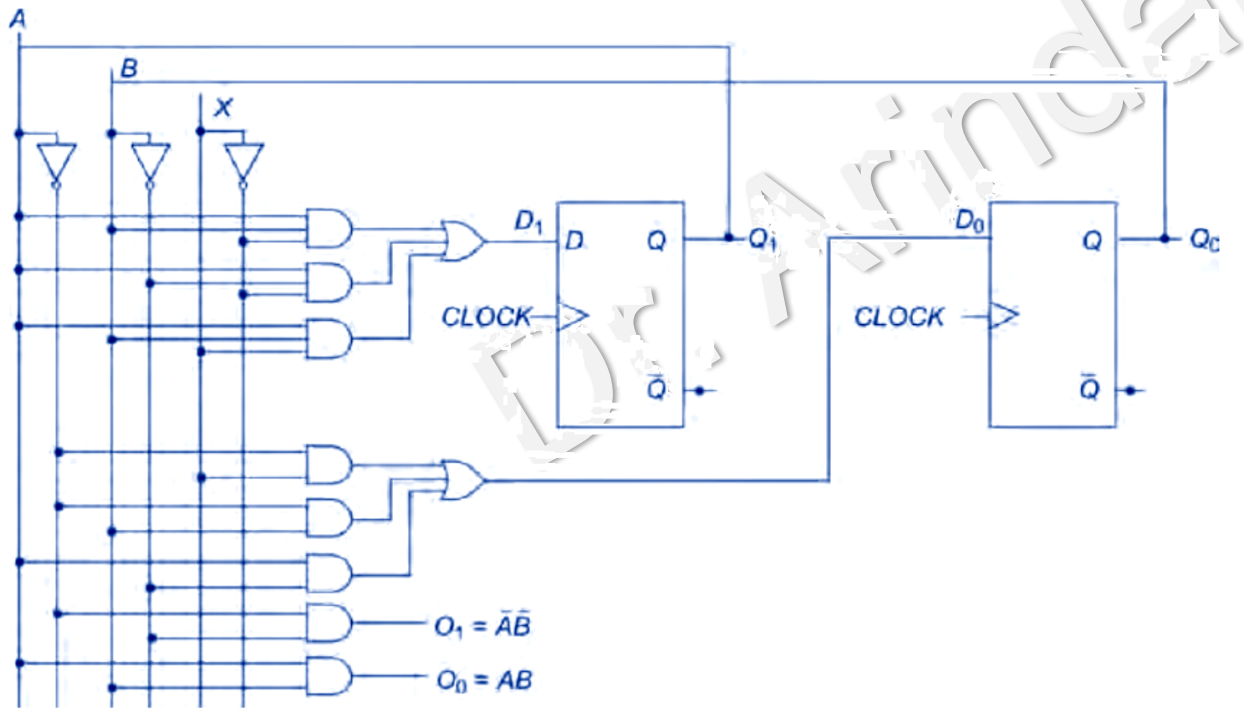


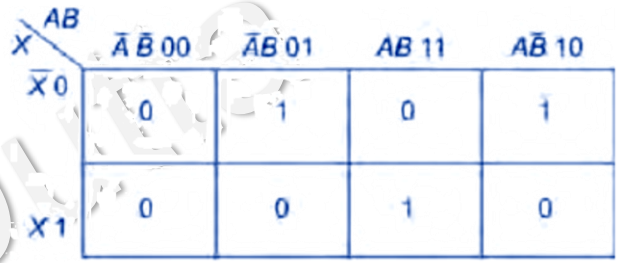
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Sequence Detector

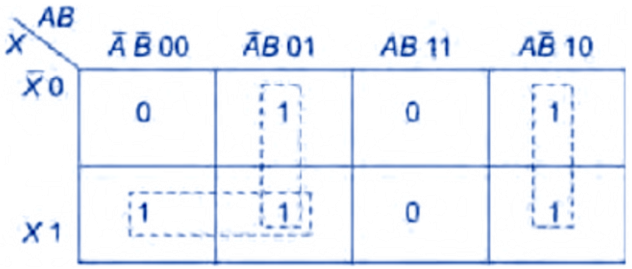
Example: Design a sequence detector as per the state table below and implement using D flip-flops.

Present state	Next state		Flip-flop inputs				Output	
	$X=0$	$X=1$	$X=0$		$X=1$		$X=0$	$X=1$
AB	AB	AB	D_1	D_0	D_1	D_0	O_1	O_0
00	00	01	0	0	0	1	1	0
01	11	01	1	1	0	1	0	0
10	11	01	1	1	0	1	0	0
11	00	10	0	0	1	0	0	1





$$D_1 = \bar{X}\bar{A}B + \bar{X}A\bar{B} + XAB$$



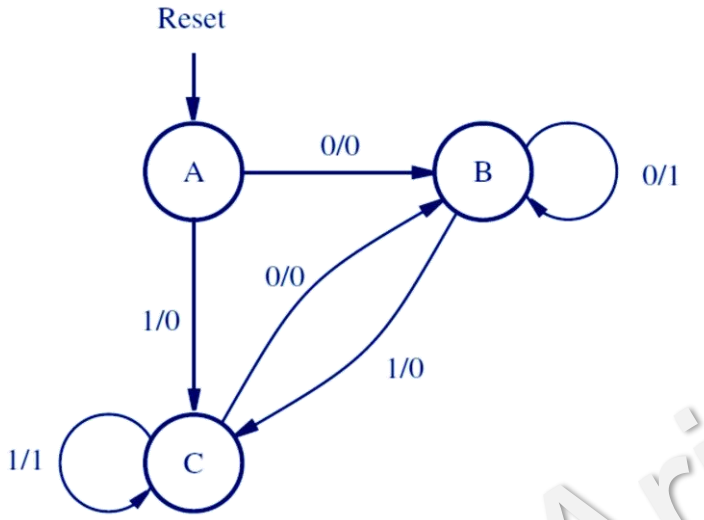
$$D_0 = X\bar{A} + \bar{A}B + A\bar{B}$$

From state table: $O_1 = A'B'$ and and $O_0 = AB$.

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Sequence Detector

Example: Design a Mealy-type FSM that has an input w and an output z. The machine is a sequence detector that produces $z = 1$ when the previous two values of w were 00 or 11; otherwise $z = 0$.



Present state	Next state		Output z	
	$w = 0$	$w = 1$	$w = 0$	$w = 1$
A	B	C	0	0
B	B	C	1	0
C	B	C	0	1

	Present state	Next state		Output	
		$w = 0$	$w = 1$	$w = 0$	$w = 1$
	y_2y_1	Y_2Y_1	Y_2Y_1	z	z
A	00	01	11	0	0
B	01	01	11	1	0
C	11	01	11	0	1

$$Y_1 = 1$$
$$Y_2 = w$$
$$z = \overline{w}y_1\overline{y}_2 + wy_2$$

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